PV-Finder: primary vertex reconstruction and track association with neural networks

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Motivation

Find the primary vertices and their tracks at the HL-LHC.

- About 200 interactions per crossing along a few cm of beam line.
- PV-Finder: a neural network turns tracks into a vertex-density histogram.
- A graph neural network (GNN) assigns tracks to the vertices.
- Compared with AMVF on the same events, with the same definitions.

$$\langle \mu \rangle = 200$$

HL-LHC, ITk

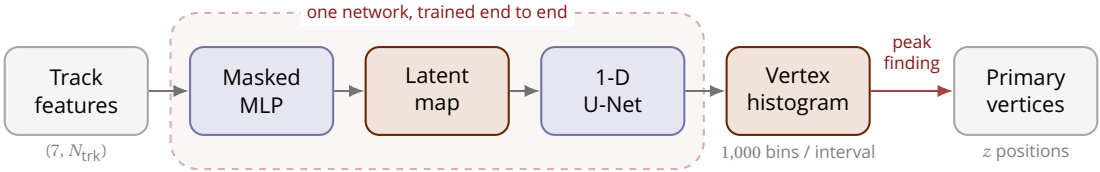
I. Method
II. Vertex finding at PU200
III. Track association and hard scatter
IV. Low pile-up and collision data
V. Timing and deployment

I

Method

Tracks, network, peaks and track association

From Tracks to a Vertex Histogram



Inputs

$d_0, z_0, \sigma_{d_0}, \sigma_{z_0}, \mathrm{cov}(d_0, z_0)$ and the interval bounds.

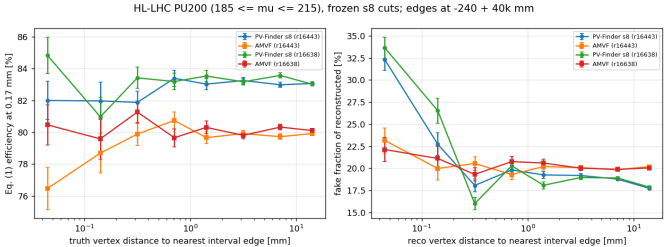
Network

Per-track MLP, summed over tracks, then a U-Net. Independent of track order.

Output

Twelve 40 mm intervals: $12{,}000$ bins of 0.04 mm over $|z| < 240$ mm.

Forty-Millimetre Intervals



Efficiency (left) and fake fraction (right) versus distance to the nearest interval edge, PU200

Why split

Fixed $1{,}000$-bin (40 mm) network output. A whole 480 mm event would need a $12{,}000$-bin output per track: $12\times$ the memory. The U-Net needs only local context.

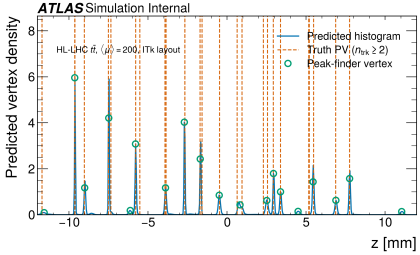
Boundaries

No overlap. Each track goes to the interval of its z_0 and carries the interval bounds as inputs. Peaks are found on the stitched $12{,}000$-bin histogram. The GNN sees the whole event.

Check

Efficiency is flat to within 0.1 mm of an edge (82–85% against 83.1%). Extra fakes at the edges: 0.2 per event, 1% of the fakes. AMVF, with no intervals, is the control.

Peak Finding and the Working Point



	HL-LHC	Low pile-up
Density network	s8	e400
Amplitude threshold	0.01	0.01
Minimum width [bins]	3	3
Minimum integral	0.184	0.50
Minimum height	0.0184	none
Position	local centroid, ± 3 bins	
GNN score cut t	0.94	0.94
Matching window	0.17 mm	
Hard-scatter window	0.1 mm	
Vertex after the GNN	≥ 2 tracks	

Two networks, two sets of cuts: integral and height are in units of each network's output, each set chosen at a matched fake rate. HL-LHC cuts at low pile-up: $+1.8$ points efficiency, but 4.0 fakes per event instead of 2.4 (AMVF 2.5).

Track-to-Vertex Association with a GNN

Track nodes: d_0, z_0, errors, θ, ϕ, p_T.

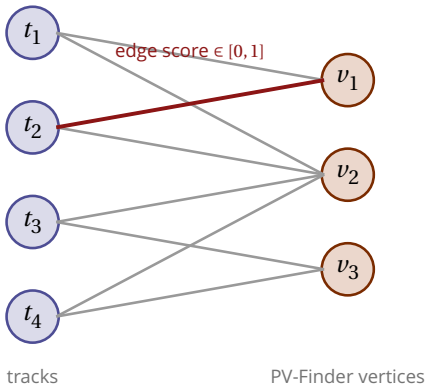
Vertex nodes: PV-Finder position and peak height.

Edges: each track to its $k = 20$ nearest vertices, with

$$\frac{z_0 - z_{\mathrm{PV}}}{\sqrt{\sigma_{z_0}^2 + \sigma_{\mathrm{PV}}^2}}, \quad \frac{d_0}{\sigma_{d_0}}, \quad |z_0 - z_{\mathrm{PV}}|.$$

Output: a score per edge. A track goes to its best vertex if the score passes $t = 0.94$.

Graph attention network, 39,521 parameters. Size not scanned yet: training and validation loss are equal and flat, and more training data helped, so a wider network is the next test.



HL-LHC Samples

$\sqrt{s} = 14$ TeV $t\bar{t}$, ITk, $|\eta| < 4$

Density network

5.17M training events at PU200.

It sees one 40 mm interval at a time, and PU200 events already contain low-density intervals ($|z| > 100$ mm).

It works at all μ: the efficiency gain over AMVF is largest at $\mu < 10$ (+4.6 points). Position at $\mu < 20$: 1–1.6 μm worse than AMVF.

GNN

149,820 training events at all μ, 0 to 200.

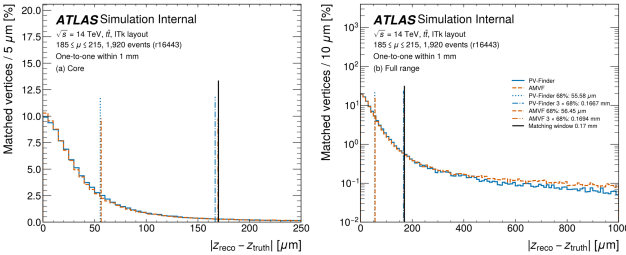
Evaluation

Held-out events at all μ, 0 to 200, never used in training.

PU200: 1,920 events at $185 \leq \mu \leq 215$, $\langle\mu\rangle = 192.5, 111.3$ truth vertices per event with ≥ 2 tracks.

Tracks: $p_T \geq 0.9$ GeV, $|d_0| \leq 1$ mm, $\sigma(d_0) \leq 0.35$ mm, $|z_0| \leq 200$ mm.

Matching Window: 0.17 mm



Single-vertex position residual

68% of matched vertices lie within:

PV-Finder	$55.6\ \mu$m
AMVF	$56.5\ \mu$m

Window $= 3 \times 56.5\ \mu\text{m} \simeq 0.17$ mm, the same for both algorithms.

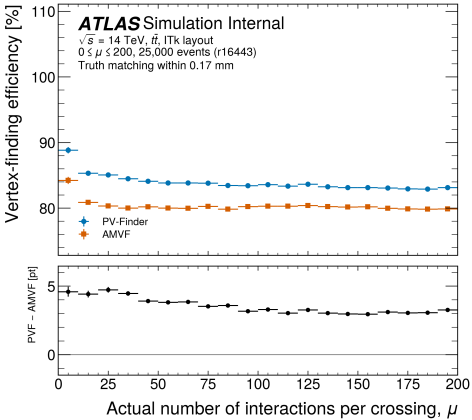
Hard scatter: 0.1 mm, as in the ITk performance studies.

II

Vertex finding at PU200

Before track association, same events as AMVF

PU200: Vertex-Finding Efficiency



Efficiency before track association

PV-Finder	83.1%
AMVF	79.9%

$$+3.20 \pm 0.06$$

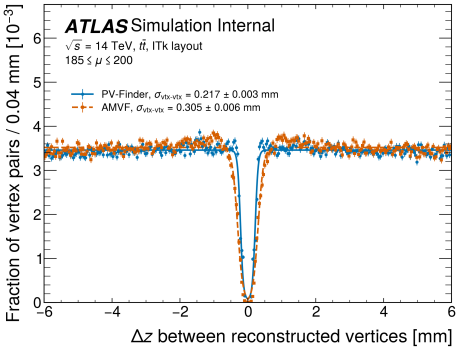
points over AMVF

Efficiency (paper Eq. 1): truth vertices with ≥ 2 tracks that have at least one reconstructed vertex within 0.17 mm, divided by all truth vertices with ≥ 2 tracks.

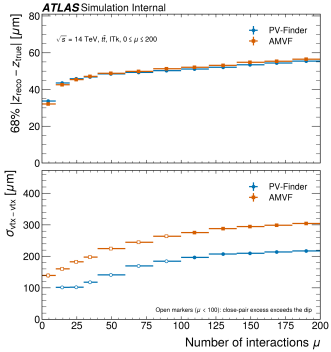
Fake: a reconstructed vertex with no truth vertex within 0.17 mm. Per event: 18.6 PV-Finder, 19.7 AMVF.

Within 0.1 mm: 74.0% vs 71.0%.

Vertex–Vertex Resolution



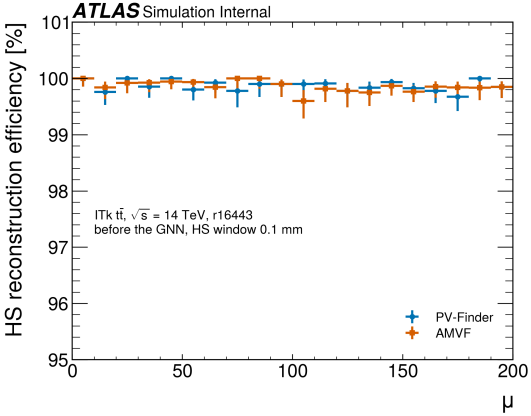
Δz between reconstructed vertices, PU200



68% residual (top) and $\sigma_{\mathrm{vtx-vtx}}$ (bottom) versus μ

PU200: $\sigma_{\mathrm{vtx-vtx}} = 0.217 \pm 0.003$ mm for PV-Finder, 0.305 ± 0.006 mm for AMVF.

Hard-Scatter Vertex Reconstruction



A reconstructed vertex within 0.1 mm of the hard-scatter vertex.

99.9%

PV-Finder at PU200

99.8%

AMVF

III

Track association and hard scatter

After the GNN: categories, hard scatter and position

Definitions after Track Association

Vertex categories, from track content

Every vertex with ≥ 2 tracks, both algorithms. Tracks are counted, not weighted. Dominant truth vertex: the one that contributes most of the vertex tracks.

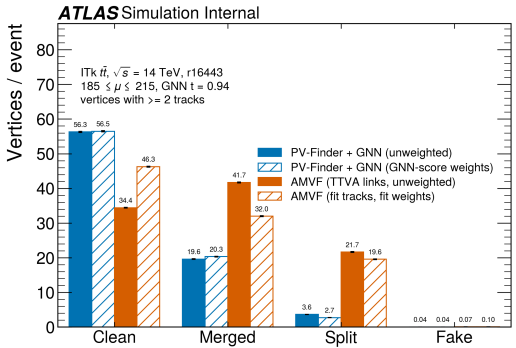
- **Clean:** $\geq 70\%$ of its tracks from the dominant truth vertex.
- **Merged:** $< 70\%$ from the dominant truth vertex.
- **Split:** another vertex with the same dominant truth vertex has a larger $\sum p_T^2$.
- **Fake:** the dominant group is tracks with no truth vertex.

Clean / truth: Clean vertices divided by truth vertices with ≥ 2 tracks.

Reference assignments to the PV-Finder vertices

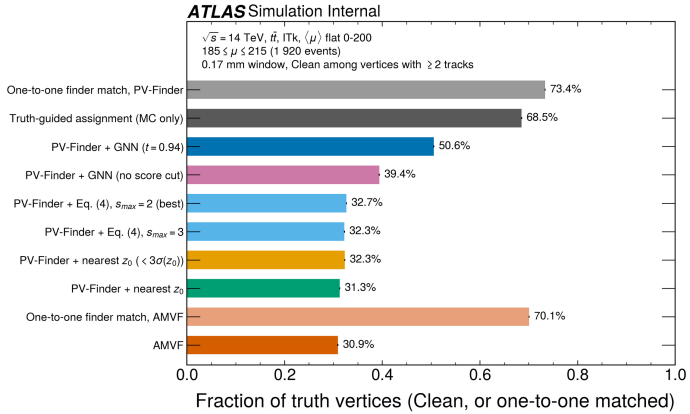
- **Nearest z_0:** each track to the vertex nearest in z.
- **z_0 within 3σ:** the same, only if $|z_0 - z_{\mathrm{PV}}| < 3\sigma(z_0)$. $\sigma(z_0)$ is the track's own z_0 uncertainty from the track fit.
- **One to one (finder cap):** vertices paired with truth vertices within 0.17 mm, closest pairs first, each used once; divided by truth vertices with ≥ 2 tracks.
- **Truth-guided (simulation only):** each track to the vertex nearest its own truth vertex, if within 0.17 mm. The best any assignment can do with these vertices.

Why AMVF Is Close to Nearest z_0



- AMVF and nearest z_0 agree within 0.4 points of Clean / truth in every μ bin.
- Both attach about 100% of the tracks, 31% wrongly at PU200, so most vertices fail the 70% rule (Merged 43%, Split 22%).
- The headline uses AMVF's TTVA links, which analyses use. With AMVF's own fit tracks and fit weights: 46.3 Clean vertices per event, 41.6% Clean / truth (TTVA links: 30.9%).
- Run 2, $\mu = 60$: our AMVF classification matches ATLAS (23.8 against ≈ 23.5 matched vertices per event). ITk publishes no categories at $\mu = 200$.
- GNN: $+8.5$ points from learned assignment, $+11$ from leaving ambiguous tracks unassigned.

Full Chain at PU200



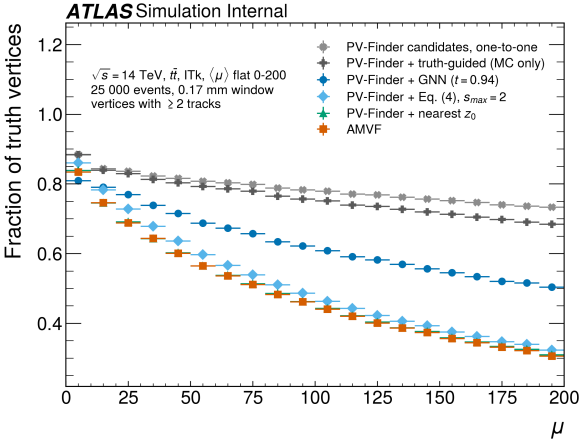
Clean vertices / truth vertices

PV-Finder + GNN	50.6%
Nearest z_0	31.3%
z_0 within 3σ	32.3%
AMVF	30.9%

Vertices with ≥ 2 tracks for every arm.

The GNN raises the clean-vertex yield $1.6\times$ over AMVF and z_0-based assignment.

Full Chain across Pile-up



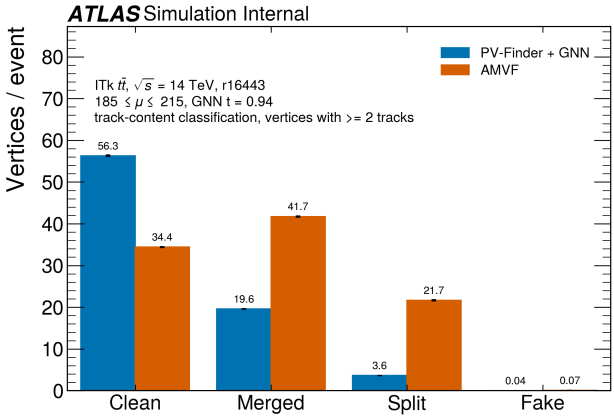
Clean vertices / truth vertices versus μ, vertices with ≥ 2 tracks for every arm.

Error bars: event bootstrap (200 replicas), smaller than the markers.

$\mu < 10$: 2.5 ± 0.6 points below AMVF. The single PU200-tuned cut t removes 11% of correctly assigned tracks where there is almost nothing to purify; small vertices drop below 2 tracks. Without the cut the GNN is 3 points above AMVF.

From $\mu = 10$ the lead grows: $+4$ points at 10–20, $+20$ at PU200. A μ-dependent t would recover low μ.

Vertex Yield by Category at PU200

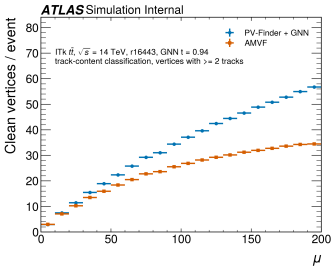


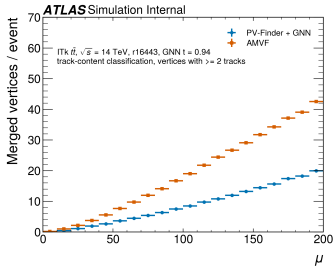
Reconstructed vertices per event

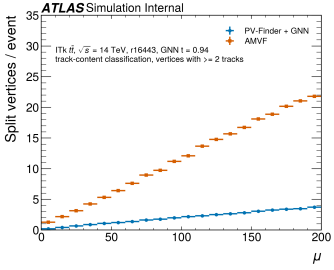
Categories as defined above, vertices with ≥ 2 tracks, the same events for both.

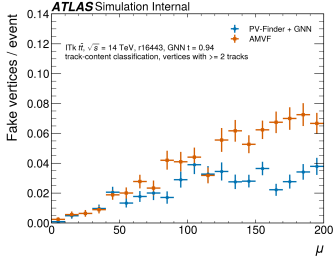
PV-Finder + GNN: more Clean vertices, about half the Merged and a sixth of the Split vertices of AMVF.

Vertex Yield across Pile-up









Clean, Merged, Split and Fake vertices per event versus μ, vertices with ≥ 2 tracks.

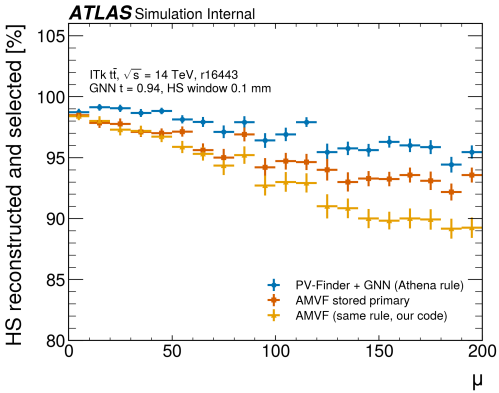
The Clean lead grows with μ. AMVF's extra vertices at high μ are Merged and Split.

Hard-Scatter Vertex after Track Association

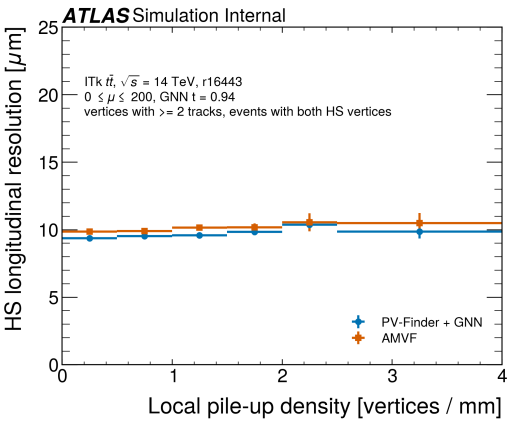
PU200	GNN	AMVF
Clean HS vertex	88.3%	92.5%
HS tracks on HS vertex	87.9%	74.1%
HS vertex purity	82.3%	88.4%
HS selected	95.1%	92.9%

HS vertex: the vertex with the most hard-scatter tracks (ATLAS truth match).
Clean HS: the vertex dominated by HS tracks is Clean ($\geq 70\%$ HS tracks).
HS tracks on HS vertex: HS tracks on it / all selected HS tracks.
HS purity: HS tracks on it / all tracks on it.
HS selected (ITk definition): the vertex with the largest $\sum p_T^2$ is the HS vertex and lies within 0.1 mm of the true HS. GNN: Athena 24.0 rule on its tracks. AMVF: its stored primary, which is the $\sum p_T^2$ leader of its fit tracks in $25{,}000$ of $25{,}000$ events.

The GNN puts more HS tracks on the HS vertex, and more pile-up tracks too: Clean HS and HS purity are below AMVF. HS selection is higher by 2.1 ± 0.5 points.



Hard-Scatter Longitudinal Resolution



What: how precisely the hard-scatter vertex is placed along z. Physics objects are built relative to it.

Residual: $z_{\text{reco}} - z_{\text{HS}}$ for the vertex closest to the true HS, within 0.1 mm.

z_{reco}**:** PV-Finder + GNN: adaptive 3D vertex fit with the beam-spot constraint on the tracks the GNN assigns to that vertex. AMVF: its own fitted vertex.

Resolution: iterative RMS (RMS within ± 3 RMS, repeated until stable), as in the ITk studies. $1{,}915$ PU200 events where both find the HS.

$\mathbf{9.5 \pm 0.3\ \mu m}$

PV-Finder + GNN

$10.2 \pm 0.3\ \mu$m

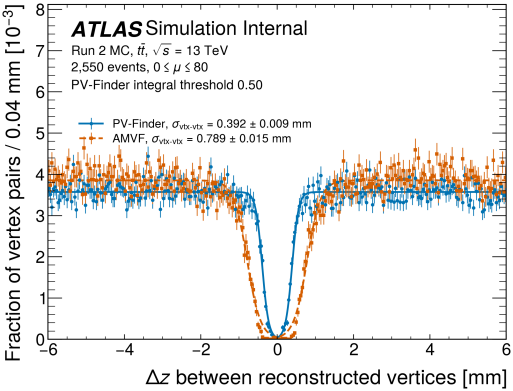
AMVF

IV

Low pile-up and collision data

Run 2 simulation, Run 2 and Run 3 data

Low Pile-up: Run 2 Simulation



$\sqrt{s} = 13$ TeV $t\bar{t}$, 2,550 test events, $\langle \mu \rangle = 38.9$

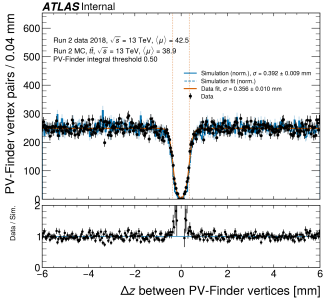
	PV-Finder	AMVF
Efficiency [%]	81.5	75.7
Fake / event	2.36	2.49
$\sigma_{\text{vtx}-\text{vtx}}$ [mm]	0.392	0.789

$$+5.8 \pm 0.1$$

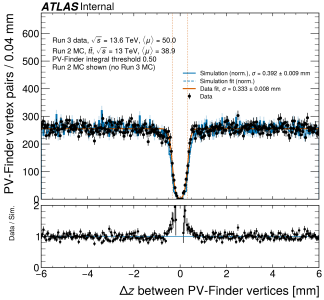
points of efficiency over AMVF

Same events, same definitions as at HL-LHC: Eq. 1 at 0.17 mm, truth vertices with ≥ 2 tracks.

Low Pile-up: Run 2 and Run 3 Collision Data



Run 2 data ($\langle\mu\rangle = 42.5$) and simulation



Run 3 data ($\langle\mu\rangle = 50.0$) and Run 2 simulation

	Run 2 simulation	Run 2 data	Run 3 data
PV-Finder $\sigma_{\text{vtx}-\text{vtx}}$ [mm]	0.392	0.356	0.333
AMVF $\sigma_{\text{vtx}-\text{vtx}}$ [mm]	0.789	0.807	0.813
AMVF vertices found by PV-Finder [%]	93.1	94.4	94.0

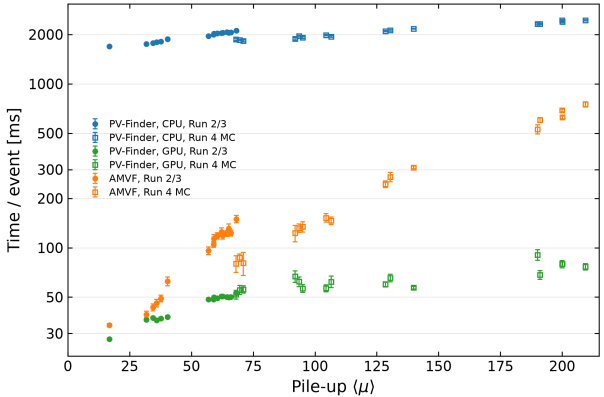
Last row: AMVF vertices with ≥ 2 tracks that have a PV-Finder vertex within 0.17 mm; in simulation too AMVF is the reference.

V

Timing and deployment

Athena integration, CPU and GPU

Athena: Timing on CPU and GPU



Same events, tracks to vertices with track links

Intel Xeon Silver 4216, NVIDIA T4.

At $\mu = 200$	ms/event
PV-Finder + GNN, GPU	76
PV-Finder + GNN, CPU	2401
AMVF, CPU	~ 650

GPU chain $\sim 9\times$ faster than AMVF.

Implemented in Athena by Trey Warner. Tools and validation in the backup.

A Smaller Network for CPU

Same data and loss, narrower U-Net

	v6	s5	s8
U-Net channels	280	112	80
CPU time [ms]	1,892	250	174
Speed-up	1	6.4×	10.9×
Δeff. at v6 fake rate [pt]		−0.18	−0.15
Δfake / event		0.01	−0.06

ONNX Runtime, one thread, one PU200 event. Small models use packed inference: tracks only, no padding, identical output.

Projected Athena time at $\mu = 200$, CPU

Chain	ms/event
v6 + GNN	2401
s5 + GNN	∼ 555
s8 + GNN	∼ 413
AMVF	∼ 650

Packed inference alone: 1.4× for v6 at no cost.
s5: 9 Nov measurement.

With the frozen small network the projected CPU chain is faster than AMVF at $\mu = 200$.

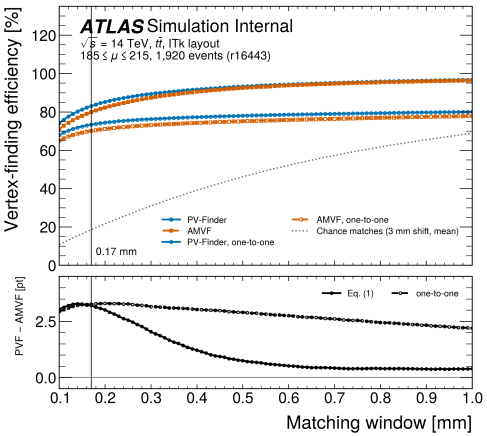
Summary

- At PU200, before track association, PV-Finder finds 83.1% of truth vertices against 79.9% for AMVF ($+3.20$ points), with fewer fakes.
- Vertex–vertex resolution 0.217 mm against 0.305 mm for AMVF.
- With GNN track association: 50.6% clean vertices per truth vertex, against 30.9% for AMVF (vertices with ≥ 2 tracks).
- Hard scatter reconstructed within 0.1 mm and selected as the highest-$\sum p_T^2$ vertex: $95.1 \pm 0.5\%$ against $92.9 \pm 0.6\%$ for AMVF (ITk definition; AMVF's own primary, consistent with the ITk paper's $\approx 92\%$). Position resolution 9.5 μm against 10.2 μm. Clean HS vertex and HS purity are below AMVF.
- Low pile-up: 81.5% against 75.7% efficiency in Run 2 simulation; $\sigma_{\mathrm{vtx-vtx}}$ about half of AMVF's in simulation, Run 2 and Run 3 data.
- Running in Athena on CPU and GPU.

Next: complete the paper draft with the frozen configuration.

Backup

Backup: Efficiency Difference versus Matching Window



PV-Finder minus AMVF efficiency, same events, for windows from 0.1 to 1 mm.

PV-Finder is ahead at every window.

Backup: Run 2 and HL-LHC Resolution

AMVF / PV-Finder $\sigma_{\text{vtx}-\text{vtx}}$: 2.01 in Run 2 simulation, 1.40 at PU200.

Track-based vertex precision: 75.6 μm (Run 2), 44.4 μm (HL-LHC).

A 0.02 mm grid moves the PU200 target from 0.137 to 0.124 mm.

Training target through the same peak finding

$\sigma_{\text{vtx}-\text{vtx}}$ [mm]	Run 2	PU200
Target	0.203	0.145
PV-Finder	0.392	0.217
AMVF	0.789	0.305
PV-Finder / target	1.9	1.5

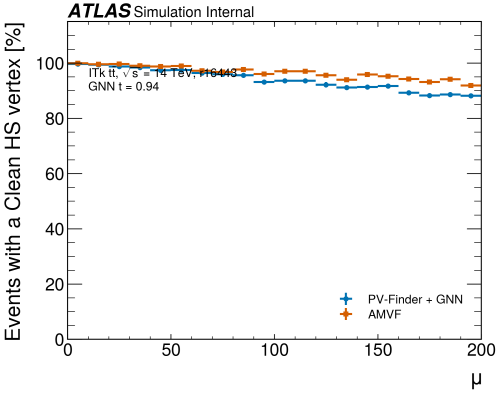
Backup: AMVF Hard-Scatter Selection

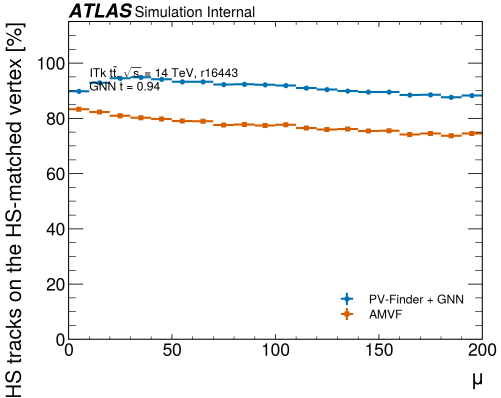
Which AMVF number is fair. Largest-$\sum p_T^2$ vertex from different AMVF track lists, compared with the AMVF stored primary, PU200:

Track list	Agrees with stored primary
Fit tracks (what Athena's sorting ranks)	100.0% ($25{,}000$ / $25{,}000$ events, all μ)
Fit tracks with weight > 0.5	97.0%
TTVA links	77.7%

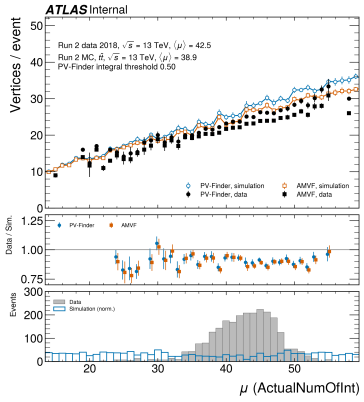
The stored primary is the Athena $\sum p_T^2$ rule on AMVF's own inputs: 92.9% HS reconstructed and selected at PU200, as in the ITk paper ($\approx 92\%$). The same rule on the TTVA links, which Athena never ranks, gives 89.1%.

Backup: Hard-Scatter Tracks and Clean HS versus Pile-up





Backup: Low Pile-up Vertex Yield and the Reference

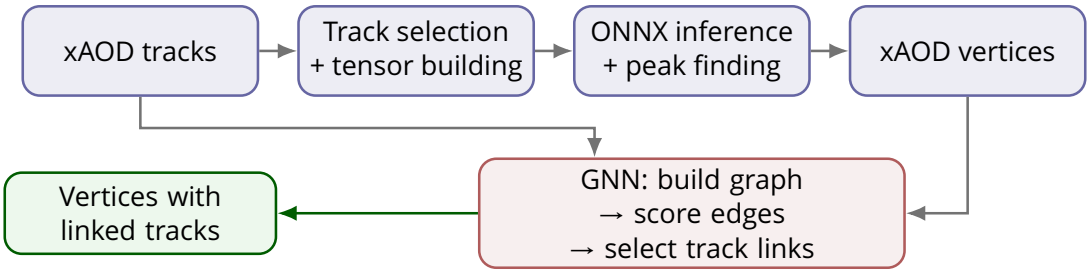


PV-Finder vertices without a reference vertex within 0.17 mm, per event

Simulation, truth reference	2.36
Simulation, AMVF reference	3.00
Run 2 data, AMVF reference	3.01
Run 3 data, AMVF reference	3.19

In data the reference is AMVF; simulation with the same reference reproduces the data.

Backup: Athena Integration



`InDetNNPriVxFinderTool` and `InDetGNNTrackVertexAssociationTool`, with the same `IVertexFinder` interface as AMVF. ONNX Runtime on CPU or CUDA, set by configuration. Athena output agrees with the Python implementation to floating-point precision.

Note Sec. 5.1 (18 Sep 2026).